DOUBLE JEOPARDY 5 categories · 400–2000 pts each

NORMAL DIST & SAMPLING

$400 Scores on a standardized test are normally distributed with mean 500 and standard deviation 100. By the Empirical Rule, approximately this percentage of test-takers scored between 400 and 600.

ANSWER **What is 68%?**

WORKED SOLUTION

400 = 500 - 100 = mu - 1 SD and 600 = 500 + 100 = mu + 1 SD. By the Empirical Rule, 68% of data falls within 1 standard deviation of the mean.

$800 A population has mean mu = 50 and standard deviation sigma = 20. For random samples of size n = 25, this is the standard error of the sampling distribution of x-bar.

ANSWER **What is 4?**

WORKED SOLUTION

Standard error = sigma / sqrt(n) = 20 / sqrt(25) = 20 / 5 = 4.

$1200 A fair six-sided die is rolled 15 times. The mean of the first 10 rolls is x-bar-1 and the mean of the remaining 5 rolls is x-bar-2. The mean of the sampling distribution of (x-bar-1 - x-bar-2) equals this value.

ANSWER **What is 0?**

WORKED SOLUTION

For a fair die, mu = 3.5. Both groups have the same population mean, so mu(x-bar-1 - x-bar-2) = 3.5 - 3.5 = 0.

$1600 For the sampling distribution of the sample proportion p-hat to be approximately Normal, both of these two conditions involving n and p must be satisfied.

ANSWER **What is np >= 10 and n(1-p) >= 10 (the Large Counts condition)?**

WORKED SOLUTION

Large Counts condition: np >= 10 and n(1-p) >= 10. When both are satisfied, the sampling distribution of p-hat is approximately Normal with mean p and SD = sqrt(p(1-p)/n).

$2000 A strongly right-skewed population has mean 100 and SD 30. This theorem guarantees that for large samples (n >= 30), the sampling distribution of x-bar is approximately Normal, regardless of the population shape.

ANSWER **What is the Central Limit Theorem (CLT)?**

WORKED SOLUTION

CLT: for random samples of size n >= 30 from any population with mean mu and SD sigma, the sampling distribution of x-bar is approximately Normal with mean mu and standard error sigma / sqrt(n).

2-VAR QUANTITATIVE DATA

$400 A data set has: mean HS GPA = 3.45, mean College GPA = 3.10, Sx = 0.40, Sy = 0.55, r = 0.68. The slope of the LSRL for predicting college GPA from HS GPA equals this value.

ANSWER **What is 0.935?**

WORKED SOLUTION

b = r x (Sy / Sx) = 0.68 x (0.55 / 0.40) = 0.68 × 1.375 = 0.935.

a = y-bar - b(x-bar) = 3.10 - 0.935(3.45) = -0.126. LSRL: y-hat = -0.126 + 0.935x.

$800 Using a regression model to make a prediction for an x-value that is far outside the range of observed data is called this.

ANSWER **What is extrapolation?**

WORKED SOLUTION

Extrapolation: using a model to predict y for x-values outside the range of the observed data. The relationship may not hold beyond the data range, making the prediction unreliable.

$1200 A single data point that has an extreme x-value and substantially changes the slope or intercept of the regression line when included is called this type of point.

ANSWER **What is an influential point?**

WORKED SOLUTION

An influential point has an extreme x-value and substantially changes the slope or intercept when included vs. excluded. Fit the LSRL with and without the point to check.

$1600 When a scatterplot shows an exponential curved relationship, a common transformation applied to the y-variable to linearize the data before regression is this.

ANSWER **What is a log transformation?**

WORKED SOLUTION

Taking log(y) often straightens an exponential relationship. After transforming, check the residual plot of the new model for random scatter to confirm the transformation worked.

$2000 A computer output shows t = 3.8 for the slope of the LSRL with p-value = 0.003. At alpha = 0.05, this is the correct conclusion about the population slope beta.

ANSWER **What is reject H0: there is convincing evidence of a linear relationship between x and y?**

WORKED SOLUTION

H0: beta = 0. p-value = 0.003 < alpha = 0.05, so reject H0. There is convincing evidence that a linear relationship exists between the variables.

INFERENCE: PROPORTIONS

$400 In the U.S., 36% of people have blood type A positive. For a one-sample z-test using 150 Norwegians, the standard deviation used in the test statistic is this.

ANSWER **What is sqrt(0.36 × 0.64 / 150)?**

WORKED SOLUTION

For a significance test, use the null hypothesis value p0 in the standard deviation: sqrt(p0(1-p0)/n) = sqrt(0.36 × 0.64 / 150) ≈ 0.039.

$800 A 95% confidence interval for the proportion of voters supporting a candidate is (0.48, 0.56). This is the correct interpretation of this interval.

ANSWER **What is: we are 95% confident the true proportion of all voters who support the candidate is between 0.48 and 0.56?**

WORKED SOLUTION

We are 95% confident the true population proportion falls between 0.48 and 0.56. The interval estimates the parameter, not individual voters.

$1200 In a sample of 150 Norwegians, 66 had blood type A positive. Before performing inference, you verify that 66 >= 10 and 84 >= 10. This is the name of the condition you just checked.

ANSWER **What is the Large Counts condition?**

WORKED SOLUTION

Large Counts condition for a confidence interval: n(p-hat) >= 10 and n(1-p-hat) >= 10. Here: 66 >= 10 and 84 >= 10. Both pass, so the sampling distribution of p-hat is approximately Normal.

$1600 Students were randomly assigned to standard or adaptive practice. The 99% CI for the difference in passing proportions (standard minus adaptive) is (0.03, 0.24) and the 90% CI is (0.06, 0.21). At alpha = 0.05, this is the conclusion of the two-proportion z-test.

ANSWER **What is reject H0: convincing evidence that the passing proportions differ?**

WORKED SOLUTION

Both CIs exclude 0. Since the 99% CI excludes 0, we reject H0 at alpha = 0.01, and therefore at alpha = 0.05 as well. There is convincing evidence that the passing proportions differ.

$2000 Researchers test H0: pD = pP vs. Ha: pD > pP at alpha = 0.01 and get p-value = 0.12. If the drug actually works, the researcher's conclusion of 'no effect' constitutes this type of error.

ANSWER **What is a Type II error (failing to reject a false null hypothesis)?**

WORKED SOLUTION

p = 0.12 > alpha = 0.01, so fail to reject H0. If the drug actually works, H0 is false, and failing to reject it is a Type II Error.

Type I Error: rejecting a true H0. Type II Error: failing to reject a false H0. Power = 1 - P(Type II Error).

INFERENCE: MEANS

$400 A 95% confidence interval for the mean weight of African bush elephants is (12.2, 14.6) thousand pounds. This is the correct interpretation of this interval.

ANSWER **What is: we are 95% confident the true mean weight of all African bush elephants is between 12.2 and 14.6 thousand pounds?**

WORKED SOLUTION

We are 95% confident the true population mean weight falls between 12.2 and 14.6 thousand pounds. The interval is about the parameter, not individual elephants.

$800 A one-sample t-test for means is conducted using a random sample of n = 30 students. The number of degrees of freedom used to find the p-value is this.

ANSWER **What is 29?**

WORKED SOLUTION

df = n - 1 = 30 - 1 = 29.

$1200 68 volunteers are randomly split into two groups: one works standing, the other sitting. After 9 months, mean back problems are compared. The most appropriate inference procedure is this.

ANSWER **What is a two-sample t-test for a difference in means?**

WORKED SOLUTION

Two separate independent groups, each receiving one treatment. The appropriate test is a two-sample t-test for mu1 - mu2. This is not matched pairs because each subject only experiences one condition.

$1600 A consumer group takes samples of 100 airplanes from each of two airlines and constructs a 95% CI for the difference in mean seat widths. If the sample size decreases to 50 per airline (all else equal), this is the effect on the confidence interval.

ANSWER **What is the interval becomes wider?**

WORKED SOLUTION

Smaller n increases the standard error, which increases the margin of error, making the interval wider. Decreasing n from 100 to 50 means less precision.

$2000 A sociologist compares salaries of bachelor's (x-bar = $55,000, s =$3,500, n = 32) and master's (x-bar = $58,000, s =$ 4,000, n = 28) degree holders. With H0: mu-M minus mu-B = 0, this is the correct formula for the two-sample t-test statistic.

ANSWER **What is t = (58,000 - 55,000) / sqrt(4,000² / 28 + 3,500² / 32)?**

WORKED SOLUTION

t = (x-bar1 - x-bar2) / sqrt(s1²/n1 + s2²/n2) = (58,000 - 55,000) / sqrt(4000²/28 + 3500²/32).

= 3000 / sqrt(571,429 + 382,813) = 3000 / sqrt(954,241) ≈ 3000 / 977 ≈ 3.07. Add variances (s²/n), not standard deviations.

CHI-SQUARE TESTING

$400 A chi-square test for homogeneity across four high schools comparing absentee rates gave a test statistic of 19.02 and p-value of 0.025. Assuming H0 states all schools have the same rates, this is the correct interpretation of the p-value.

ANSWER **What is: assuming absentee rates are the same for all four schools, there is a 2.5% probability of observing a test**

statistic of 19.02 or larger?

WORKED SOLUTION

The p-value assumes H0 is true. If absentee rates are equal across schools, a test statistic of 19.02 or larger would occur only 2.5% of the time. Since $0.025 < 0.05$, reject H0.

$800 In a chi-square test, the formula used to calculate each cell's contribution to the overall test statistic is this.

ANSWER **What is (Observed - Expected)^2 / Expected?**

WORKED SOLUTION

chi^2 = sum of $(O - E)^2 / E$ over all cells. Expected count = (row total x column total) / grand total.

$1200 A chi-square test of independence is conducted using a 4 by 3 contingency table. The degrees of freedom for this test equal this value.

ANSWER **What is 6?**

WORKED SOLUTION

$df = (\text{rows} - 1)(\text{columns} - 1) = (4 - 1)(3 - 1) = 3 \times 2 = 6$.

$1600 Before conducting a chi-square test, you must verify that all expected cell counts meet this minimum threshold.

ANSWER **What is all expected counts must be at least 5?**

WORKED SOLUTION

Large Counts condition for chi-square: all expected cell counts must be at least 5. If this fails, combine categories or collect more data.

$2000 A researcher surveys 200 randomly selected people, recording political affiliation (Dem/Rep/Ind) and stance on a policy (Support/Oppose). The null hypothesis for the appropriate chi-square test states this.

ANSWER **What is: there is no association between political affiliation and policy stance in the population?**

WORKED SOLUTION

This is a chi-square test of independence. H0: no association between political affiliation and policy stance. Ha: there is an association. df = (3-1)(2-1) = 2.